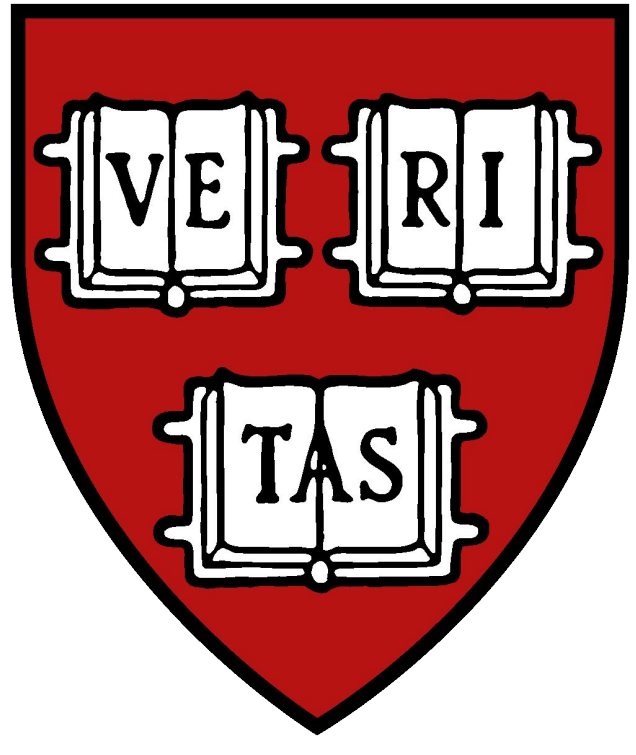


Beam-Climbing Robot Controlled by a Soft Ring Oscillator



Arthur K. MacKeith¹, Won-Kyu Lee², George M. Whitesides^{2,3,4}

¹The University of Chicago, Chicago, IL

²Department of Chemistry and Chemical Biology, Harvard University, Cambridge, MA

³Wyss Institute for Biologically Inspired Engineering, Boston, MA

⁴Kavli Institute for Bionano Science and Technology, Harvard University, Cambridge, MA
Department of Energy, Award # DE-SC0000989



Introduction

Three of the main strengths of soft robots are their low cost, their simplicity, and their ability to deform. All three of these advantages are undermined when soft robots rely on hard components to generate the control signals required for complex tasks such as locomotion.

An on board soft pneumatic ring oscillator is capable of supplying the pressure required for undulating locomotion while conserving the robot's overall softness. Using a new semi-soft design for a low cost ring oscillator we are able to drive robust locomotion along a beam and perform tasks like cleaning.

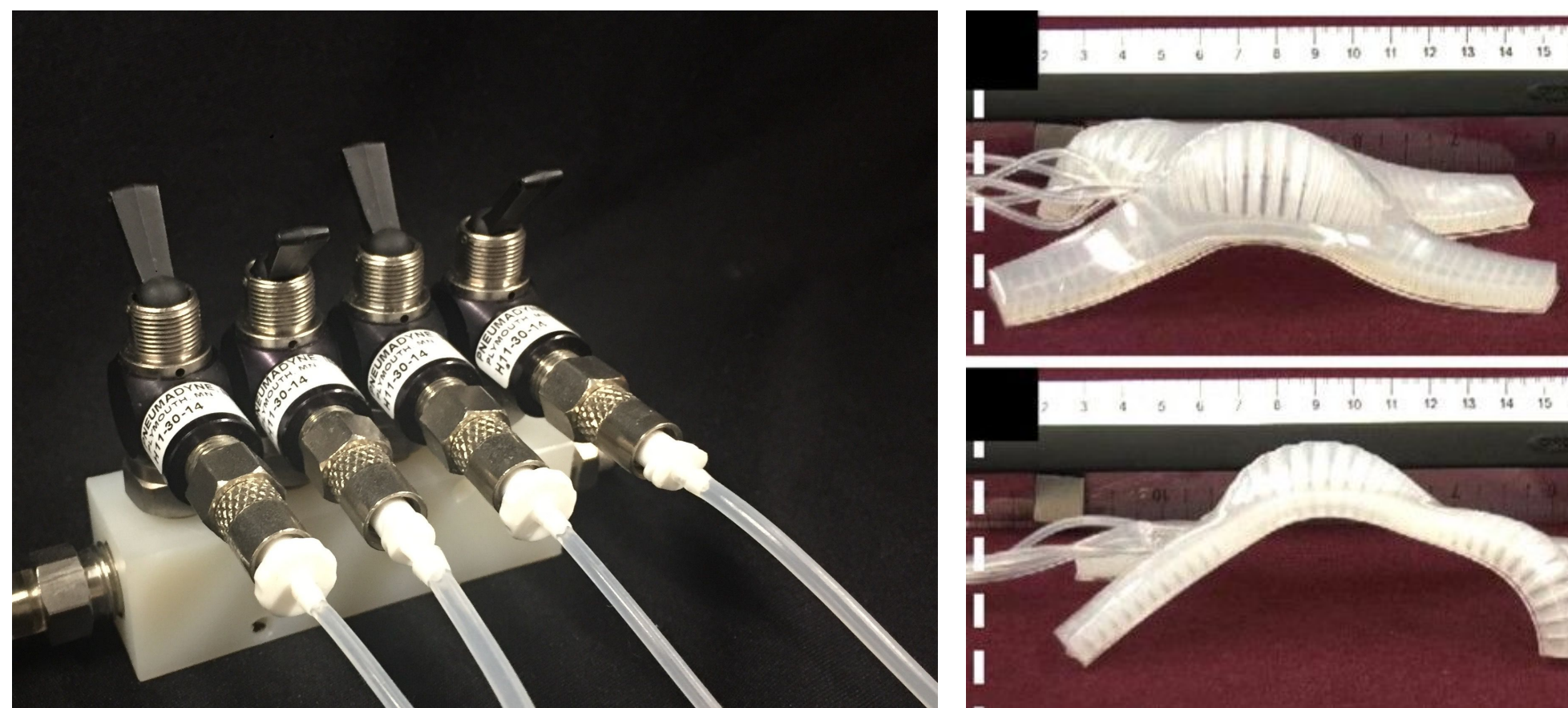


Figure 1: Hard controls and a soft robot

A beam climbing robot can find practical use in applications such as infrastructure maintenance where exposed beams frequently require painting and cleaning whose supplies and environments lend themselves to cheap disposable robots rather than human workers.

Shepherd et al., PNAS 2011, 108(51)

Methods

We constructed a Buckling Ring Oscillator (BRO) from semi-soft materials that are deformable but inextensible. This device is made from three pneumatic inverters, such that, as long as a supply pressure is provided, when a high pressure is supplied a low pressure is output and visa versa. When three of these are connected in a ring there is no stable configuration. Therefore each inverter continuously oscillates, each one with a different time offset.

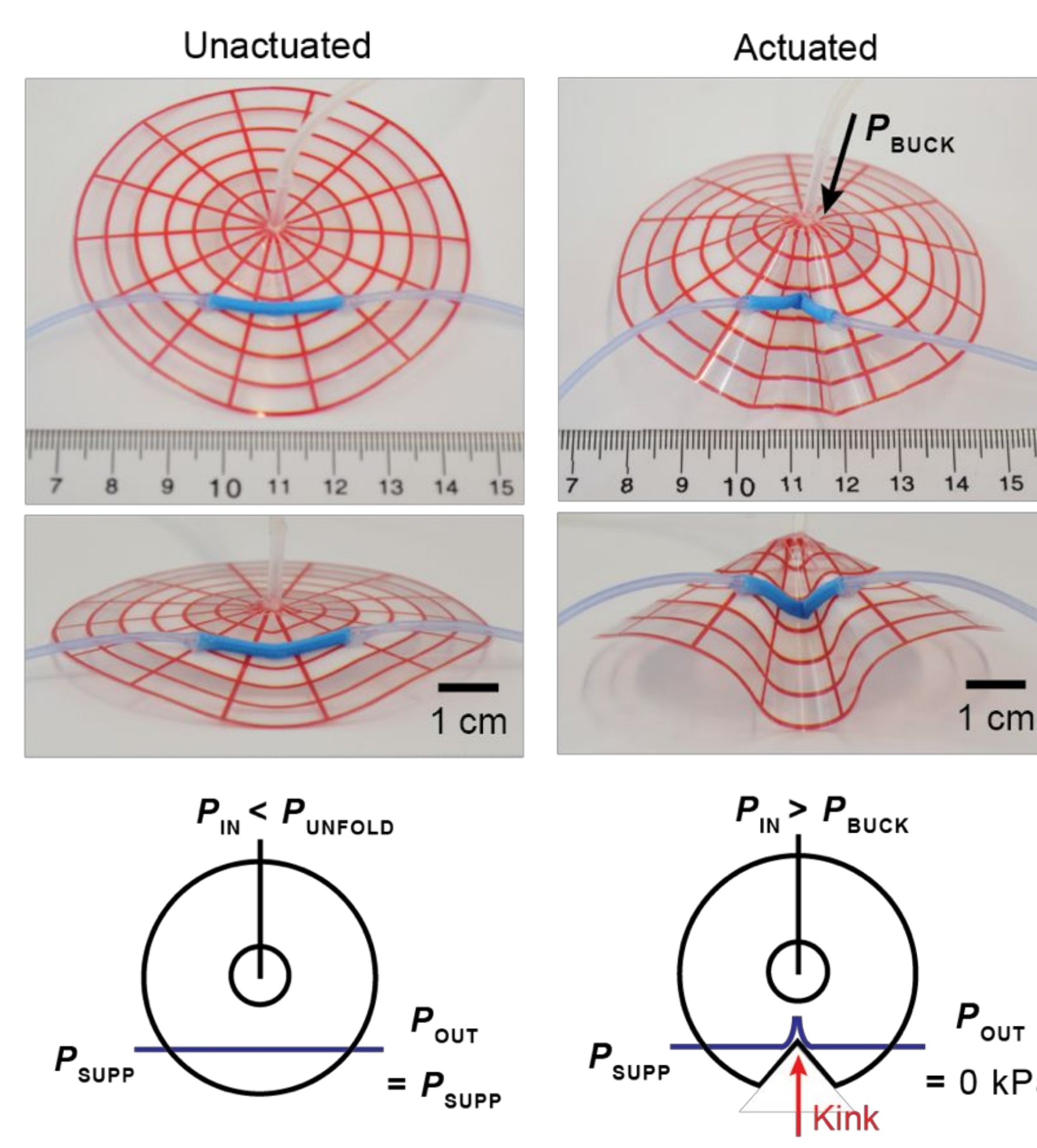


Figure 2: Pneumatic inverter

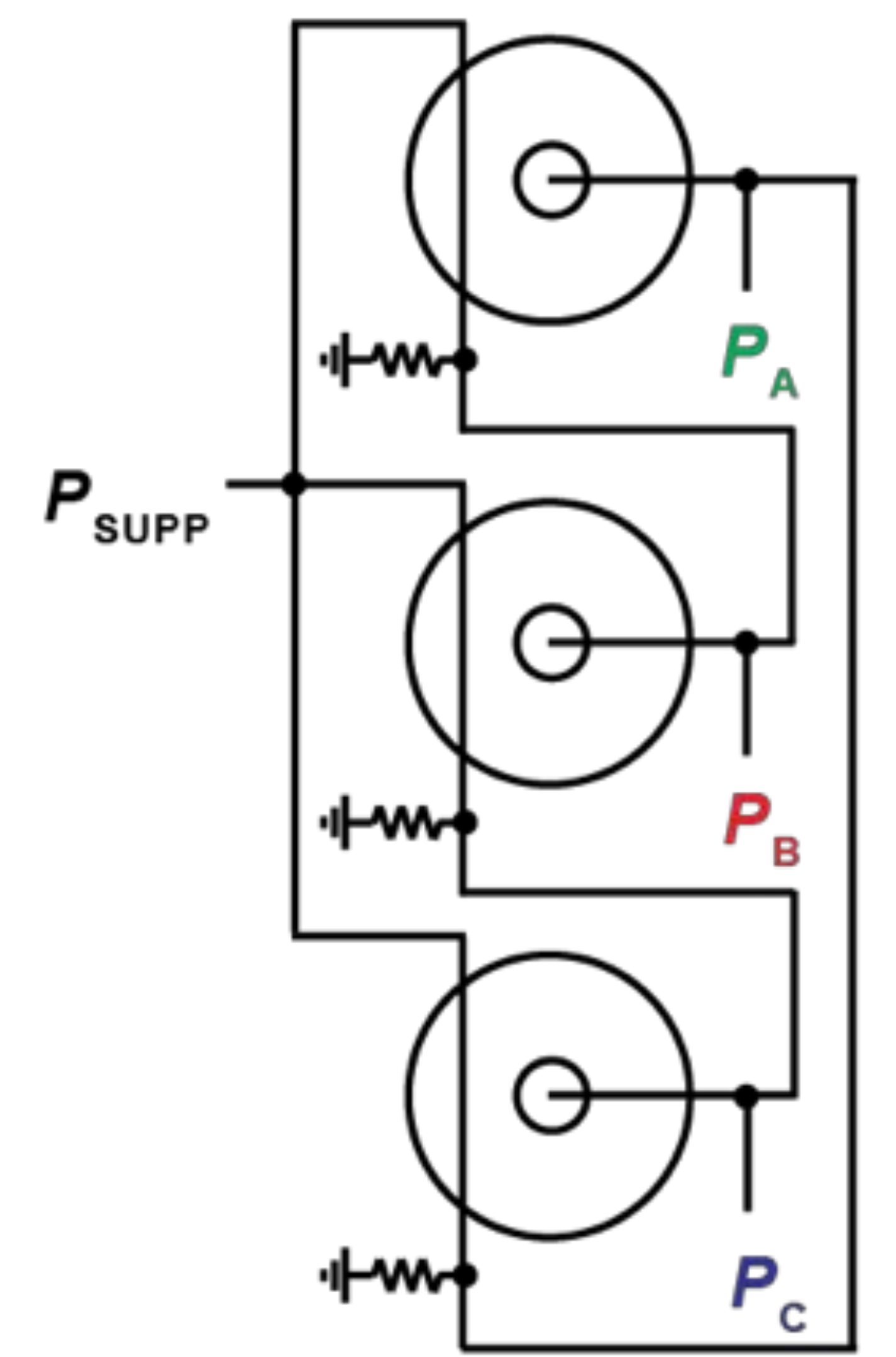


Figure 3: Buckling Ring Oscillator (BRO) schematic

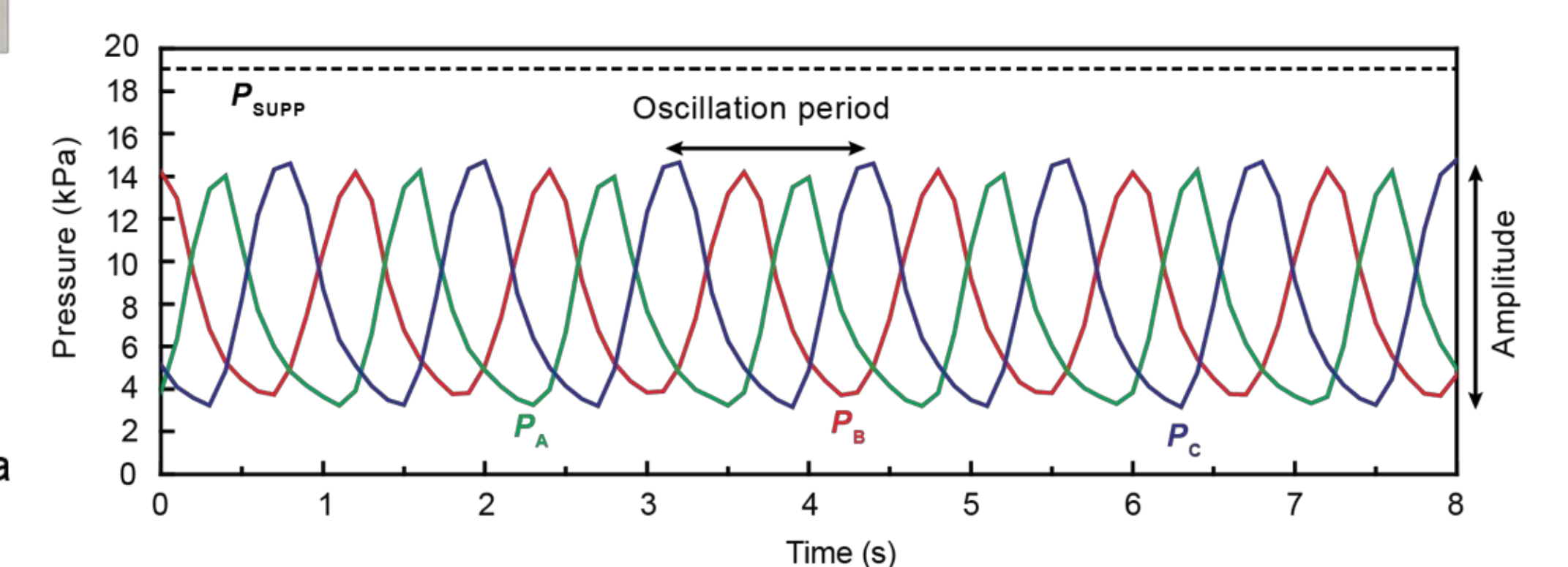
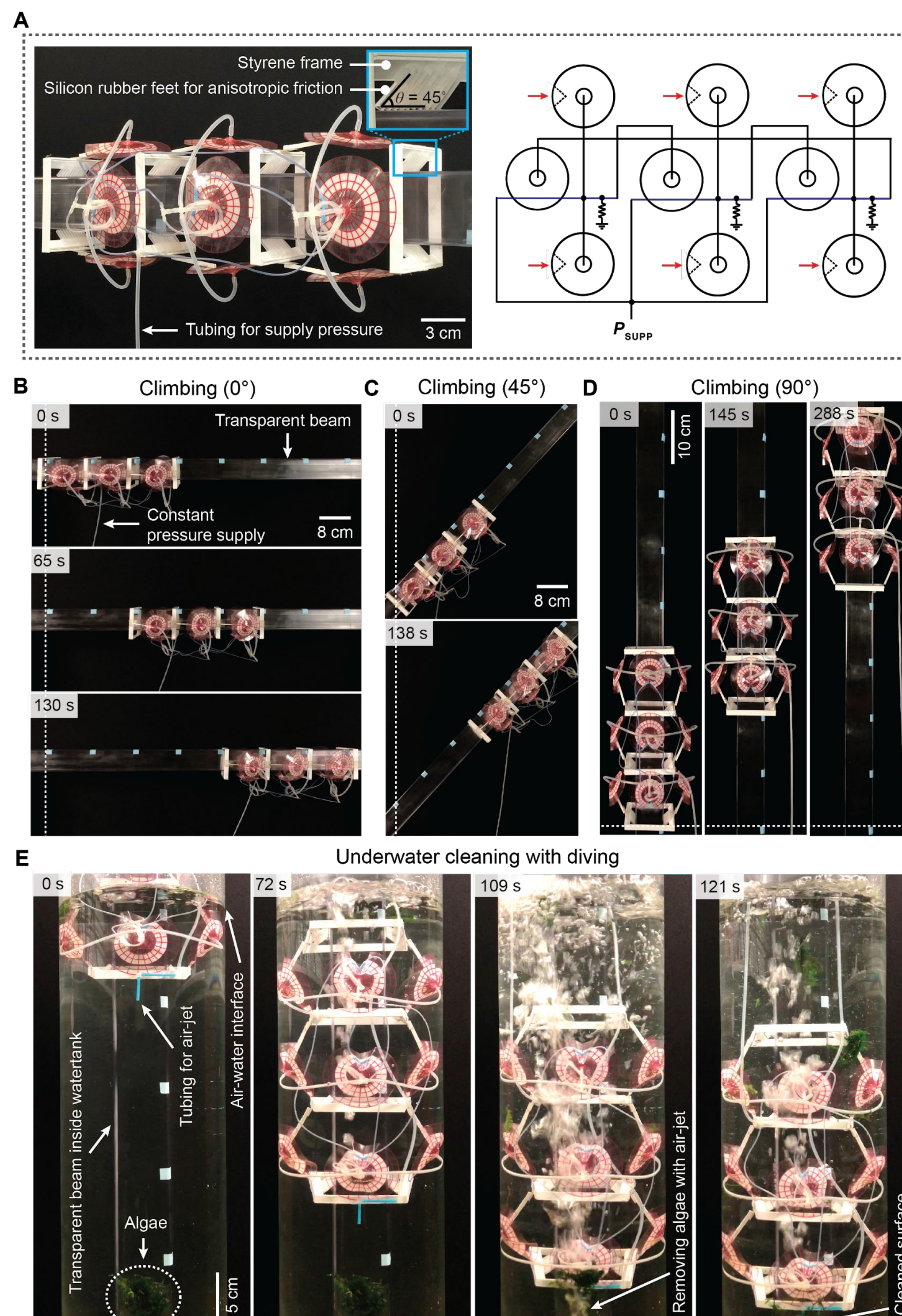


Figure 4: Buckling Ring Oscillator (BRO) pressure signals

Lee et al., In preparation

Results

The BRO was used as a signal generator to power locomotion along a beam. Two additional terminated inverters were attached to each of the inverters in the



BRO to be used as actuators. These actuators are mounted between styrene frames that grip the beam with elastomeric rubber feet. These feet make a 45° angle with the beam creating anisotropic friction, such that the undulations of the BRO and actuators achieve motion along the beam. Unlike prior soft and semi soft climbing robots this one does not require any off board control components, rather it only requires a single constant pressure source to achieve locomotion.

The low cost of this robot should also be noted, the materials used to construct it cost less than \$5 and are industrially available. Furthermore, the only specialized equipment used to make it was a laser cutter. Its low cost means that this design would be well suited to tasks like cleaning or painting exposed beams when one might dispose of the device after use.

Figure 5: The beam-climber robot was fabricated by supplementing a BRO with two additional inverters and a three pairs of semi-soft feet (A). Horizontal motion with a speed of 18 cm/min (B), 13 cm/min at 45° (C), and 10 cm/min upwards (D). The beam-climber can also dive underwater against its own buoyancy with a constant speed of 20 cm/min and perform underwater cleaning with an air-jet attached to the front of the robot (E).

Lee et al., In preparation

Conclusion

This work demonstrates how complex tasks like locomotion can be achieved by substituting hard, offboard control components, for inexpensive, commercially available semi-soft components.

It also shows that it could be used to address real world problems like external maintenance on structures (e.g. bridges that require frequent painting and cleaning of their beams).

Acknowledgments

This work was funded by Department of Energy Award # DE-SC0000989. Thanks to Haihui Joy Jiang, Dan Preston, Christoffer Abrahamsson, Markus Nemitz, Amit Nagarkar, Vanessa Sanchez, Jeff Rawson and Sam Root.

Contact

Arthur K. MacKeith: amackeith@uchicago.edu
Won-Kyu Lee: wlee@gmwhgroup.harvard.edu
George M. Whitesides: gwhitesides@gmwhgroup.harvard.edu

